

Types of network

While there are many ways to classify networks, two common criteria are size and topology (layout). A small home network is organized very differently from a global network, such as the internet.

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Size

The size of a network is the physical distance between its nodes. The most common network sizes are local area networks (LANs), metropolitan area networks (MANs), and wide area networks (WANs). Distance affects the number of routers needed, the kind of media used to connect the nodes, and the type of information shared over the network.

LAN (local area network)

A LAN is a small wired or wireless network. Typically, all the devices are in a single building, or even on a single floor in a large building. Too many computers in a LAN can lead to delays in sending and receiving information.

MAN (metropolitan area network)

A MAN is a network that covers a city. Essentially, it's any network bigger than a LAN but too small to be considered a WAN. Large university campuses sometimes have MANs with fibre optic connections between buildings.

WAN (wide area network)

A WAN is a network that covers more than 48 km (30 miles). It connects devices using copper wire, satellites, or fibre optic cables. Big companies, such as Google, Microsoft, and Facebook, need WANs because they have offices in different cities. The internet is the largest WAN in existence.

